

EduSphere: Technical & System Documentation

Comprehensive Architecture, AI Engine, Blockchain & Operational Reference

Institutional Profile: P.E.S. Modern College of Engineering • **System Version:** 2.0-Production
Core Stack: Full-Stack Micro-Monolith (React 19, Python Flask 3, MongoDB, Groq LPU Cloud, Algorand TestNet)

1. Executive Summary & Problem Formulation

Campus placement operations involve coordinating hundreds of students across multiple engineering branches with dozens of corporate partners. Conventional workflows suffer from fragmented spreadsheets, absence of early predictive indicators for at-risk students, lack of automated preparation tooling (such as ATS resume checks and realistic mock technical interviews), and unverified academic credentials.

EduSphere addresses these challenges through four foundational pillars:

- Real-Time Placement Intelligence:** High-throughput MongoDB aggregation pipelines calculate branch placement rates, salary distribution histograms, skill demand frequencies, and gender equity metrics.
- Generative AI Student Suite:** Powered by Groq Cloud (Llama 3.3 70B & GPT-OSS-120B), offering ATS resume OCR, adaptive 20-question mock tests, voice-based multi-lingual mock interviews (ARIA), and career roadmaps.
- Algorand Blockchain Anchoring:** Every student record modification is cryptographically signed and anchored onto the Algorand TestNet in a 0-ALGO note transaction, creating an immutable public verification trail.
- Digital Twin Campus:** An interactive 3D WebGL model of the college campus alongside a simulated 2D IoT sensor network for crowd density analytics and anomaly detection.

2. System Architecture & Technology Stack

Layer	Technologies & Frameworks	Key Role & Responsibility
Frontend UI	React 19.2, Vite 5.4, React Router v7	Single-page reactive interface, role-based navigation, responsive views
Design & UX	Vanilla CSS3 Glassmorphism tokens, Chart.js	Premium dark-mode aesthetic, responsive interactive charts & toast feedback
Speech & 3D	Web Speech API, Three.js WebGL	ARIA multi-lingual voice loop (11 langs) & 3D campus digital twin
Backend API	Python 3.10+, Flask 3.0, Gunicorn WSGI	RESTful Blueprints, CORS, keep-alive heartbeat, file upload handling
Database	MongoDB Atlas, PyMongo 4.6	Flexible document store, aggregation pipelines, indexed lookups
AI Inference	Groq Cloud SDK (Llama 3.3 70B, GPT-OSS-120B)	High-speed LPU inference, structured JSON output, OCR parsing
Blockchain	Algorand TestNet (py-algorand-sdk 2.8)	Tamper-proof academic credential anchoring via zero-ALGO transaction notes
DevOps	Docker Multi-Stage, Kubernetes (k8s), Render	Production containerization, zero-downtime automated deployment

3. Database Schema & Data Models

Collection	Primary Schema Fields	Indexes & Constraints
users	_id, username, email, password (hashed), role ('admin' 'student'), student_id, created_at	Unique index on email (students), unique username (admin)

students	_id, name, email, branch, cgpa, skills[], projects, internships, placed, resume_text, gender, blockchain_tx_id, created_at	Indexed on branch, placed, and case-insensitive email
companies	_id, name, industry, min_package, max_package, requirements[], website, created_at	Indexed on name and industry classification
placements	_id, student_id (FK), company_id (FK), role, package (LPA), placement_date, status ('confirmed' 'pending')	Compound index on student_id + company_id, sorted by date
test_history	_id, student_id (FK), test_title, difficulty, subject, score, total_questions, answers[], questions[], completed_at	Indexed on student_id, sorted descending by completed_at

4. Core Functional Modules

4.1 Student Experience & Generative AI Tools

- **ARIA (Voice-First AI Technical Interviewer):** Conducts adaptive, multi-turn technical and behavioral mock interviews. Features an animated reactive CSS avatar with talking rings and voice bars, speech recognition (STT) and natural speech synthesis (TTS) across 11 languages/accents (English US/IN, Hindi, Marathi, Tamil, Telugu, Kannada, Gujarati, French, German, Spanish). Evaluates candidate emotion (Confident, Nervous, Enthusiastic, Unsure) and produces a comprehensive final performance dossier with scored dimensions.
- **AI Resume ATS Analyzer & OCR:** Accepts both digital PDFs (via PyPDF2) and scanned documents/images (via Tesseract OCR). Extracts candidate skill sets, calculates match scores against target job profiles, flags formatting deficiencies, and suggests actionable enhancements.
- **Dynamic 20-Question Mock Test Generator:** Generates syllabus-aligned multiple-choice questions across difficulty levels (Beginner, Intermediate, Advanced) and subjects (DSA, General Aptitude, Core CS, Full-Stack). Includes a countdown timer, answer review mode, detailed rationale explanations, and persistent score tracking in MongoDB.
- **Predictive Salary Calculator & 6-Month Roadmap:** Estimates market-accurate compensation ranges based on CGPA, verified skill count, completed projects, and internships calibrated to Indian IT hiring data. Generates milestone-driven 6-month preparation roadmaps.

4.2 Placement Officer (TPO) Intelligence Portal

- **Executive KPI Analytics:** Real-time counters for total student body, placed student count, aggregate placement percentage, average package, highest package, and total registered corporate recruiters.
- **Salary Distribution Histogram:** MongoDB \$bucket aggregation grouping confirmed student offers into market bands: 0-3 LPA, 3-5 LPA, 5-8 LPA, 8-12 LPA, 12-20 LPA, and 20+ LPA.
- **Branch-wise Placement Statistics:** Multi-stage \$lookup computing branch strength, placed counts, placement percentage, and average CTC per engineering department (CS, IT, Mechanical, etc.).
- **Skill Demand Matrix & Velocity Trends:** Aggregates top 15 skills demanded across placed students and maps monthly recruitment seasonality.
- **Data Management & CSV Export:** Instant search, branch filtering, pagination, and one-click export for students_export.csv and placements_export.csv.

4.3 Algorand Blockchain Credential Verification

To establish immutable, tamper-proof academic and placement verification, EduSphere anchors student records to the **Algorand TestNet**:

1. **Payload Serialization:** When a student is created or updated, student credentials (CGPA, skills, branch, placement status) are serialized into a compact JSON string.
2. **Cryptographic Transaction:** The backend signs a 0-ALGO self-transaction utilizing the institution's Algorand account, embedding the payload in the note field.
3. **Consensus Confirmation:** The transaction is confirmed on the Algorand TestNet within 3.5 seconds, returning a permanent transaction ID (tx_id).
4. **Public Auditing:** External recruiters can independently verify the credential timestamp and data on any public Algorand explorer without requiring internal database access.

4.4 3D Virtual Campus & 2D Cohort Analytics

- **3D WebGL Digital Twin:** Interactive architectural model of P.E.S. Modern College of Engineering featuring G+4 floor isolation, 3D exploded assembly view, cross-sectional cutter, day/night lighting, and room inspection.
- **2D Cohort Sensor Analytics:** Simulated IoT sensor network streaming live room-by-room student occupancy and automated crowd bottleneck alerts. Integrated with **Grok CampusAI**, a voice-enabled assistant that answers real-time occupancy and safety queries.

5. Comprehensive API Specification

Endpoint	Method	Access	Functional Purpose & Payload
/api/auth/login	POST	Public	Admin authentication via username & password
/api/auth/student/login	POST	Public	Student login; auto-provisions user on first visit if student email exists
/api/auth/verify	GET	JWT	Validates JWT Bearer token and returns active identity claims
/api/students	GET / POST	Auth / Admin	Paginated student query/filter; POST anchors record to Algorand TestNet
/api/students/<id>	GET / PUT / DEL	Auth / Admin	Student CRUD; updates refresh the Algorand blockchain transaction anchor
/api/companies	GET / POST	Auth / Admin	Manage recruitment partner profiles, salary bands, and skill criteria
/api/jobs/recommendations/<id>	GET	JWT	Heuristic matching engine correlating student skills with company criteria
/api/placements	GET / POST	Auth / Admin	Placement records with joined student and company document lookups
/api/analytics/stats	GET	JWT	Summary KPIs: placement percentage, avg/min/max packages, total counts
/api/analytics/salary-distribution	GET	JWT	Histogram ranges (0-3, 3-5, 5-8, 8-12, 12-20, 20+ LPA)
/api/analytics/branch-stats	GET	JWT	Departmental breakdown of student counts, placed counts, and average CTC
/api/ai/analyze-resume	POST	JWT	Multipart upload for PDF/Image OCR and ATS compatibility analysis
/api/ai/interview	POST	JWT	ARIA multi-turn voice interview loop, scoring, and comprehensive dossier
/api/ai/generate-test	POST	JWT	Generates 20-question dynamic MCQ mock placement test via Groq LLM
/api/ai/save-test-result	POST	JWT	Persists student exam score, chosen answers, and question set to MongoDB
/api/ai/ask	POST	JWT	Grok CampusAI sensor-aware campus occupancy query engine
/api/admin/export/students	GET	Admin	Exports entire student directory as CSV attachment (students_export.csv)
/api/ping	GET	Public	Keep-alive ping endpoint preventing cloud container sleep

6. Production Deployment & Local Setup

- **Docker Multi-Stage Container:** Stage 1 (Node 18 Alpine) compiles the React frontend (npm run build). Stage 2 (Python 3.10 Slim) installs system tesseract-ocr, copies backend requirements, and bundles the compiled frontend into frontend-react/dist, serving via Gunicorn WSGI on port 5001.
- **Kubernetes Ready:** Complete production manifests in k8s/ (deployment.yaml with health probes, service.yaml, ingress.yaml, and secrets.yaml).
- **Render Cloud Platform:** Configured via render.yaml for zero-downtime deploys with automatic 10-minute client keep-alive pings.

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